DYNAMIC PROGRAMMING

Problem 1 : Playing with Numbers

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*
1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format
First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input
6

Sample Output

6

Source Code :

```
#include <stdio.h>
long long countWays(int n) {
    if (n == 0 || n == 1 || n == 2) {
        return 1;
    }
    long long w_minus_3 = 1;
    long long w_minus_2 = 1;
    long long w_minus_1 = 1;
    long long current_ways = 0;
```

```
    for (int i = 3; i <= n; i++) {
        current_ways = w_minus_1 + w_minus_3;
        w_minus_3 = w_minus_2;
        w_minus_2 = w_minus_1;
        w_minus_1 = current_ways;
    }
    return current_ways;
}
int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        printf("%lld\n", countWays(n));
    }
    return 0;
}
```

Output :

Input	Expected	Got
6	6	6
25	8641	8641
100	24382819596721629	24382819596721629

Problem 2 : Playing with chessboard

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:
Input
3
1 2 4
2 3 4
8 7 1
Output:
19

Explanation:
Totally there will be 6 paths among that the optimal is
Optimal path value:1+2+8+7+1=19

Input Format
First Line contains the integer n
The next n lines contain the n*n chessboard values
Output Format

Print Maximum monetary value of the path

Source Code :

```
#include <stdio.h>
#include <stdlib.h>
long long max(long long a, long long b) {
    return (a > b) ? a : b;
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    long long *dp = (long long *)malloc(n * sizeof(long long));
    long long temp;
    for (int j = 0; j < n; j++) {
        scanf("%lld", &temp);
        if (j == 0) {
            dp[j] = temp;
```

```
        } else {
            dp[j] = dp[j-1] + temp;
        }
    }
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%lld", &temp);
            if (j == 0) {
                dp[j] += temp;
            } else {
                dp[j] = temp + max(dp[j], dp[j-1]);
            }
        }
    }
    printf("%lld\n", dp[n-1]);
    free(dp);
    return 0;
}
```

Output :

Input	Expected	Got
3 1 2 4 2 3 4 8 7 1	19	19
3 1 3 1 1 5 1 4 2 1	12	12
4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28

Problem 3 : Longest Common Subsequence

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Source Code :

```
#include <stdio.h>
#include <string.h>
int max(int a, int b) {
    return (a > b) ? a : b;
}
int lcs(char *s1, char *s2) {
    int m = strlen(s1);
    int n = strlen(s2);
    int dp[2][n + 1];
    for (int j = 0; j <= n; j++) {
        dp[0][j] = 0;
        dp[1][j] = 0;
    }
    for (int i = 1; i <= m; i++) {
        int bi = i & 1;
        for (int j = 1; j <= n; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                dp[bi][j] = 1 + dp[1 - bi][j - 1];
```

```
            } else {
                dp[bi][j] = max(dp[1 - bi][j], dp[bi][j - 1]);
            }
        }
    }
    return dp[m & 1][n];
}
int main() {
    char s1[1000], s2[1000];
    if (scanf("%s", s1) != 1) return 0;
    if (scanf("%s", s2) != 1) return 0;
    printf("%d\n", lcs(s1, s2));
    return 0;
}
```

Output :

Input	Expected	Got
aab azb	2	2
ABCD ABCD	4	4

Problem 4 : Longest non-decreasing Subsequence

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Source Code :

```
#include <stdio.h>
#include <stdlib.h>

int upper_bound(int *arr, int low, int high, int target) {
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] <= target) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int *seq = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &seq[i]);
    }
    int *tails = (int *)malloc(n * sizeof(int));
    int len = 0;
    for (int i = 0; i < n; i++) {
```

```
        int idx = upper_bound(tails, 0, len, seq[i]);
        tails[idx] = seq[i];
        if (idx == len) {
            len++;
        }
    }
    printf("%d\n", len);
    free(seq);
    free(tails);
    return 0;
}
```

Output :

Input	Expected	Got
9 -1 3 4 5 2 2 2 2 3	6	6
7 1 2 2 4 5 7 6	6	6